compounds, and map terrain to support site planning for mining operations or construction activities. Their aerial or micro-propulsion mobility allows them to reach locations inaccessible to rovers or human teams, including crater interiors, steep cliffs, lava tube entrances, and asteroid surfaces with irregular gravitational fields.

Drones will act as essential tools for early-stage resource evaluation, guiding where to deploy robotic miners, excavation devices, and processing units. In low-gravity and vacuum environments such as asteroids, drones equipped with ion thrusters or cold-gas microjets can provide precise maneuvering for material sampling, anchoring, and surface stabilization tasks. On the Moon and Mars, autonomous drones may support the construction of habitats, radiation shelters, solar power arrays, and communication relay towers, using regolith-based 3D printing and coordinated robotic assembly. They can also assist with transport of small payloads and spare components, extending logistical reach across scattered work sites.

In addition to physical operations, drones play a crucial role in realtime monitoring and maintenance of extraterrestrial infrastructure. They can inspect mining equipment for mechanical wear, monitor dust accumulation on solar panels, survey structural stability of habitats, and detect hazardous environmental changes such as radiation spikes or regolith shifting. Over time, networks of cooperative drones may form autonomous industrial ecosystems, coordinating with ground robots and orbital platforms to maintain continuous productivity without constant human oversight.

By enabling efficient resource extraction, adaptive construction, and operational resilience, drones will be instrumental in laying the foundation for long-duration scientific bases, commercial mining sites, and eventual human settlements in space. Their capabilities transform them from